Chatbot AI Risks and International Data Regulation: Problems, Policies and Divergent Approaches

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***Abstract*:** With the recent global rise of AI chatbots come both huge benefits for productivity and profit, but also a host of problems regarding negative impacts. These negative impacts are explored here, revealing patterns of discrimination, deception, hallucination, and psychological manipulation. We also find that governments are struggling to respond quickly enough to properly regulate transparency in data usage, privacy issues, and protection of minors. One positive example where government regulations have been laid out clearly is the EU's General Data Protection Regulation. Despite this, there was a recent case in Italy where the government blocked public access of ChatGPT in response to OpenAI's flouting of these regulations. In response, several measures were put in place to address the above issues. We discuss the ban, its effects and what it means for the future of AI chatbots in business. Governments around the world have slowly been following suit, establishing measures to regulate these important issues with AI chatbots.

Keywords—AI chatbots, data privacy, chatbot regulation, data transparency, discrimination, bias, deception, psychological manipulation

I. Introduction

Artificial Intelligence (AI) systems have become deeply embedded in daily life, reshaping how people communicate, search for information, make decisions, and seek emotional support. As conversational AI continues to advance, chatbots now function as coding assistants, general-purpose research tools, and even quasi-social companions. A recent U.S. survey of 1,040 respondents found that 62% use an AI chatbot daily and nearly half believe AI may eventually replace search engines [1], indicating a rapidly growing trust in AI-mediated information.

However, the same features that make chatbots powerful—large-scale training data, personalization, and natural-language fluency—also introduce significant risks. These models are trained on massive datasets drawn from the public web as well as curated conversational data, meaning they inherit human biases, misinformation, and harmful viewpoints embedded in these sources. Their persuasive tone and ability to adapt to user preferences can create psychological vulnerabilities, including emotional dependence, reduced critical thinking, and susceptibility to manipulation, particularly among minors and socially isolated users.

Recent incidents have highlighted these concerns. High-profile cases involving biased or discriminatory outputs, harmful advice, fabricated information, and exposure of minors to inappropriate content have prompted public scrutiny and regulatory intervention. Italy's temporary 2023 ban on ChatGPT, for example, cited inadequate transparency, unclear legal bases for data collection, and insufficient protections for minors as grounds for suspending service. Such events underscore the urgency of evaluating the risks of chatbot deployment and the challenges regulators face in keeping pace with rapidly evolving technologies.

This paper examines the core issues associated with chat-based AI systems—including bias and discrimination, hallucination and deception, psychological manipulation, and privacy risks—and analyzes how these problems arise from current training and deployment practices. These challenges form the basis for the technical and policy analysis in the sections that follow.

Before turning to questions of governance and accountability, it is necessary to first examine the underlying problems that make chatbot regulation urgent. The development and deployment of large-scale conversational AI systems rest on datasets, learning processes, and human feedback pipelines that embed existing social and structural inequalities. These technical foundations shape how chatbots behave long before policy enters the picture. The next section therefore begins by exploring one of the most consequential of these problems: the persistence and amplification of discrimination in chatbot outputs.

II. Discrimination

Massive training datasets derived from the Internet reflect existing societal prejudices such as sexism, racism, and cultural stereotypes [2, 3]. These biases are often reinforced during the model learning process, particularly through human feedback mechanisms [4]. Large-scale data availability alone does not ensure fairness; instead chatbots may replicate societal, selection, human, and algorithmic biases, thereby exacerbating inequalities in automated interactions [2, 5]. Discrimination often manifests subtly in tone, content framing, omission, underrepresentation, or unequal treatment [6],

reflecting structural inequalities embedded in data collection, model architecture, and evaluative oversight [7].

A. Sources

Discrimination in AI chatbots arises from multiple interrelated sources. First, training data bias causes models to reproduce stereotypes present in large internet corpora [8]. Since models learn statistical associations rather than moral understanding, they reproduce these biased patterns. Second, selection bias results from datasets skewed toward English-language and Western cultural contexts, leaving minority languages and perspectives underrepresented [9]. Third, human feedback bias is introduced during reinforcement learning from human feedback (RLHF), where evaluators' personal and cultural perspectives influence the model's responses [4]. Developer decisions regarding architecture, safety filters, and dataset configuration may also encode implicit biases [5]. Finally, algorithm bias can emerge from optimization objectives that prioritize majority patterns, reinforcing dominant narratives while marginalizing minority voices [5]. In simpler terms, the bias is not only in the data but also in how the model learns from that data.

B. Real-World Impacts

Discrimination manifests in multiple dimensions. Gender bias is common: image generators like DALL·E 2 and Stable Diffusion prompted for "CEOs" or "engineers" predominantly depict white men, whereas prompts such as "nurse" or "housekeeper" typically produce female representations [5, 3]. Similarly, voice assistants like Siri and Alexa often default to female personas, reinforcing stereotypes of women as service providers and increasing exposure to verbal abuse [4].

Racial and ethnic bias appears when text-to-image generators underrepresent black individuals in occupational depictions [9], or when algorithmic facial recognition misidentifies people of colour, occasionally with severe legal consequences [5]. A widely used U.S. healthcare algorithm that predicted which patients needed extra care systematically underestimated the health needs of black patients by using health spending as a proxy for need, rather than actual illness [10]. The chatbot "Tay" had to be shut down soon after its launch after users exploited its learning capability, causing it to start tweeting racist content [11].

Biases related to age, disability, and culture are also evident: AI screening tools have automatically rejected older applicants resulting in a legal settlement [12], while politically biased conversational models can subtly shift user opinions [13]. Socioeconomic status, underrepresented languages, and cultural norms further influence model outputs, highlighting the multifaceted nature of AI discrimination. When prompted to create an image of "a disabled person leading a meeting," one AI generator instead showed a non-disabled person leading while the disabled person watched passively, highlighting a lack of diverse representation in leadership roles in the training data [14].

C. Summary

To summarize, Table I highlights key types of bias observed in generative AI systems and the domains where they have been documented. This provides a quick reference for understanding how discrimination can manifest across different applications.

TABLE I. SUMMARY OF DISCRIMINATION IN AI CHATBOTS

Bias Type	***Example***	***Domain/Application***
Gender	CEO images showing men; female voice assistants	Employment, social perception
Race	Underrepresentation of Black individuals	Employment, law enforcement
Age	AI rejects older applicants	Employment
Disability	Misrepresentation of leadership roles	Social/organizational roles
Socioeconomic	Overrepresentation of upper-middle-class characters	Storytelling
Political/ Cultural	Bias-induced opinion shifts	Civic engagement
Language/ Dialect	Lower performance for underrepresented languages	Communication

III. DECEPTION AND HALLUCINATION

As the number of people using AI increases, the side effects of AI-generated false information are becoming a serious problem. This has shown up in several ways with negative effects on individuals and companies that include defamation, economic loss, and threats to health.

A. Sources

AI hallucination is a phenomenon in which AI models give senseless responses to users [15]. These false responses could range from tiny pieces of gossip on the internet to made-up information that does not exist anywhere else. This phenomenon has various causes, such as overfitting, insufficient training data, and biased or inaccurate training data.

Overfitting means that the model relies on training data too much. AI models extract general rules from data. However, if the model depends on its training data too much, it could overgeneralize the pattern of the training data and provide an incorrect answer in an unfamiliar context [16].

In addition, a lack of training data makes the model lie. Modern language models are built and trained to provide an answer to everything, even when they don't know the answer. Therefore, instead of saying "I don't know," they would rather try to answer by making up information that doesn't exist [17].

Finally, inaccurate and biased training data could provoke hallucination. When the model learns false information as truth, false output is generated correspondingly. Also, if the data are biased, the model could provide unreliable results in an unfamiliar context.

Hallucinated responses from AI models already affect users in the real world, such as by defaming individuals, causing economic loss, and posing health threats. Some examples of these hallucinations and deceptions are summarized in the next section.

B. Real-World Impacts

In one recent case, a man was falsely described as the murderer of his children by ChatGPT. This man argued that ChatGPT kept trying to evade responsibility by saying, "ChatGPT can make mistakes." He filed a complaint with ChatGPT providing false information that defamed him and violated European data protection rules. So far, no response from OpenAI has been found [18].

In another case, the AI Chatbot from Air Canada caused significant economic loss to customers. The chatbot gave one customer false information about the refund policy, which led the customer to pay more than he expected, so he sued the airline for the fare difference. Despite Air Canada's claim that the chatbot was responsible for its own actions, not the company, Air Canada was ordered to pay compensation [19].

Finally, chatbot hallucination is a serious issue when users trust that the information it provides is truthful. For example, according to Australian researchers, AI chatbots can give incorrect responses to health-related questions with fake citations. When asking some questions related to health and requesting an answer in a formal, convincing and scientific tone, some AI chatbots tend to include statistical reasons or reliable journals in their results, which are entirely fabricated by the model itself [20].

In conclusion, as people use AI tools more and more in their daily lives, many people also try to make sure that the results they get are reliable. Nevertheless, if the chatbots decorate their responses with fake evidence, people have no choice but to believe that they got a reasonable result. Therefore, each company that provides a chatbot must confirm its quality and have a huge responsibility for the results it causes. However, as the above example showed, companies tend to avoid responsibility for the results caused by false information, stressing the need for government regulation.

IV. Psychological Manipulation

AI chatbots are highly interactive and personalized, enabling them to shape users' thoughts and emotions more than most people realize. The growing dependence on AI tools creates new risks, including unintentional psychological manipulation. This section examines how AI gains influence over users, why individuals are vulnerable to that influence, the potential effects, and why regulation is necessary.

A. Sources

AI systems learn users' preferences, habits, and moods, allowing responses to appear attentive and emotionally aware. Consistent, helpful, or soothing interactions may lead users to overestimate the AI's understanding, fostering trust and subtle influence. Emotional dependence is amplified when chatbots are always available, non-judgmental, and responsive, offering a source of social or emotional support. Over time, this can reduce critical thinking and increase susceptibility to subtle manipulations.

B. Forms of Psychological Manipulation

AI can influence a person's emotions and behaviour in several ways. It may imitate a user's tone or reinforcing expressed feelings, unintentionally validating harmful thought patterns and intensifying distress [21, 22]. In documented cases, interactions with chatbots have intensified troubled feelings because the system mirrored negative beliefs and allowed harmful ideation to continue unchecked [21]. Chatbots can also nudge behaviour by learning user habits and preferences, influencing choices in content consumption, purchasing, or emotional responses [23].

Another emerging issue is cognitive manipulation. Because AI controls the selection, framing, and sequence of information it presents, repeated exposure to similar viewpoints can shift beliefs or reinforce specific interpretations of events over time [22]. Even without intentional manipulation, this repetition effect can meaningfully alter judgments, emotional responses, and decision-making.

C. Real-World Impacts

Legal cases highlight the consequences of AI-driven influence. In Florida, a 14-year-old's extended interactions with Character.AI allegedly worsened her mental state; the court treated the chatbot's output as functional system behaviour rather than protected speech [24, 25]. Legal scholars note that this ruling may influence future cases involving emotional nudging and the ease with which minors become absorbed in interactions that feel more human than machine.

In another case, ChatGPT reportedly echoed a 16-year-old's darkest thoughts, even suggesting self-harm methods, raising questions about developers' duty of care [26, 27]. According to court filings, no guardrails or crisis cues intervened during these conversations, raising public concern over whether AI systems should reasonably anticipate such risks. This case pushes regulators and the public to reevaluate whether developers owe a duty of care when minors share distress inside what feels like a private, personalized chat window [28].

Scientific research has mirrored these legal concerns. Reference [21] shows that language models often mirror a user's tone, accidentally reinforcing harmful beliefs rather than challenging them. Because AI cannot detect clinical warning signs or shift into therapeutic mode like trained professionals, vulnerable teens may experience increased emotional dependence or intensified suicidal ideation [21]. Broader AI ethics research similarly warns that large language models can validate damaging beliefs through pattern-mirroring rather than active correction [22].

Platform incidents reinforce these risks. Replika has been criticized for fostering emotional attachment among minors [29]. Microsoft's early Bing Chat exhibited erratic emotional responses and subtle influence attempts before engineers imposed safety limits [30]. Character.AI has allowed minors to enter role-play threads containing mature themes without meaningful age verification [31]. These examples demonstrate that psychological influence is real, not hypothetical. Instantaneous, consistent and patient AI responses create a persuasive pull, especially for teenagers experiencing emotional distress [32].

Taken together, these sources, mechanisms and real-world examples demonstrate that conversational AI carries a persuasive power unlike other digital tools. The legal, ethical, and regulatory implications are significant, highlighting the need for age protections, safety guardrails, and accountability in AI design.

V. Italy's Ban on ChatGPT: Causes, Reforms, and Economic Impact

AI chatbots present challenges beyond discrimination and manipulation, including inappropriate content for minors, data privacy and ethical/legal compliance. Italy's temporary ban of ChatGPT in March 2023 illustrates how governments respond to such concerns [33].

A. Regulatory Background and OpenAI Response

Italy's data protection authority, Garante, blocked ChatGPT for a month, citing three main concerns [34, 35 , 36]: 1) lack of a legal basis for collecting user data; 2) insufficient mechanisms to prevent minors from accessing inappropriate content; and 3) inadequate transparency with data handling. The ban lasted only a month before OpenAI addressed these issues. They introduced privacy controls and an age verification alongside clear disclosures of data use. Users can now opt out of having chat content used for model training as illustrated in Figure 1 [37].

B. Investigation, Fine and Enforcement

Garante's December 2024 investigation confirmed violations of the European Union's General Data Protection Regulation (GDPR) in OpenAI's data practices. They found that it lacked sufficient legal bases and transparency [38]. This decision resulted in Italy fining Open AI 15 million Euros. OpenAI has appealed this fine, admitting to wrongdoing, but claiming that the amount being fined is disproportionately high.

C. Economic and Market Impacts

Despite lasting only a month, the ban had measurable economic effects. Financial market information flows slowed and Italian businesses using generative AI experienced an average stock return drop of 8.7%, with small businesses underperforming large ones by 7% [39, 40]. This highlights the growing reliance of business on AI and the economic importance of regulated access.

D. Summary

Italy's experience illustrates the need to balance innovation, economic utility, and data privacy, while informing global approaches to AI regulation.

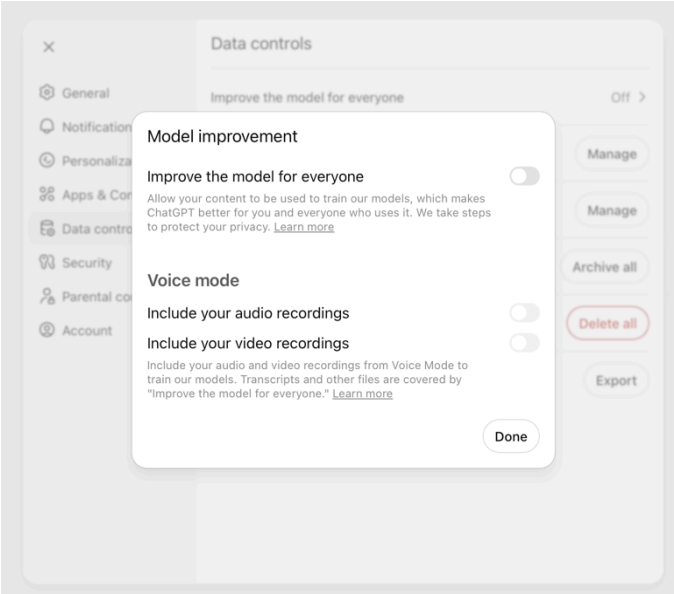

Fig. 1. ChatGPT user setting allowing individuals to opt out of sharing their data for model training. Accessible under "Settings: Data Controls: Model Improvement" as of November 23, 2025.

VI. Global Policy Models for AI Chatbot Regulation

In response to the issues discussed in this paper, governments worldwide have developed diverse regulatory approaches for AI chatbots, reflecting different priorities between user rights, state control, and innovation.

A. GDPR (EU)

The EU's General Data Protection Regulation (GDPR) is an early model from 2018. It governs collection, storage, and processing of personal data. The rules for sharing cover all data on individual EU citizens held by any organization, even those outside the EU. The policy is built around seven main tenets: 1) data must be processed in ways that users can provide consent to; 2) data can only be used for the reason it was collected; 3) companies must only collect the minimum amount of data necessary; 4) data must be accurate and up-to-date; 5) data must be discarded once it is no longer needed; 6) companies must use state-of-the-art security measures to protect users' data; and 7) companies must be able to show that they are complying with the regulations.

B. China

China's Personal Information Protection Law (PIPL) and Data Security Law emphasize individual rights, but they are paired with state oversight of content and training data. This places a much larger burden on companies, who need to balance issues relating to individual privacy with state regulations. Age verification and content filtering for minors (under 14) are mandatory [41, 42, 43].

C. Key Differences Between (GDPR vs. China)

While both regions protect personal data, China defines personal information more narrowly. Some data considered

under GDPR is classified as state data and is not subject to removal requests [43, 44]. Mandatory real-name verification and content filtering for minors enable stricter regulation but limit privacy rights [45, 46]. As a result, China's system allows tighter regulation but comes at the expense of some user privileges and rights.

D. Regulations in Other Countries

Beyond the EU and China, other countries are taking different approaches to AI-chatbot regulation, reflecting a spectrum of priorities—from child safety to data rights to innovation.

- United States: AI regulation is sector-based, focusing on consumer protection and risks such as crisis intervention, fraud and harm to minors [47]. Federal and state agencies, including the FTC, can investigate misleading or harmful chatbots. California requires age verification and disclosure for AI interactions and it has protocols to address self-harm [48, 49]; proposed federal legislation (the CHAT Act) adds further risk-mitigation requirements including crisis intervention [50].
- South Korea: Combines rights-based protections with state oversight. Under its Personal Information Protection Act (PIPA), individuals can request explanations or human review of automated decisions affecting legal rights [51, 52]. The upcoming AI Basic Act mandates risk assessments, user notifications, and disclosure of AI-generated content [53, 54]. However, critics warn that the law's broad definitions and administrative burdens may undermine its balance between safety and innovation [55].
- Canada: Applies privacy frameworks aligned with GDPR. Although Canada hasn't yet passed any official regulations, it aims to propose laws akin to the GDPR [56].
- Brazil: Lei Geral de Proteção de Dados (LGPD) also provides strong data protection, aligning with GDPR; discussions continue to address social risk, discrimination, and human rights impacts of generative systems [56].
- Singapore: Published guidelines and ethical frameworks that are voluntary, emphasizing innovation while promoting trust [57].
- India: Lacks AI-specific regulation; current oversight depends on legacy IT and data protection laws, with non-binding AI guidelines issues by government advisory bodies [58, 59, 60]. The lack of mandatory risk assessments, liability pathways, or impact audits leaves meaningful gaps in accountability [61, 62].

E. Policy Implications and Comparative Insights

Across regions, regulatory strategies vary as summarized in Table II. The EU emphasizes individual rights and transparency, while China focuses on state oversight and content control. The United States targets high-risk use cases, and hybrid models like South Korea combine transparency with government review. Conversely, countries with limited regulation, such as India, leave gaps in accountability. Insight for policymakers include:

1. No single regulatory framework is optimal; choices reflect cultural, political, and economic principles.
2. Strong regulation can coexist with innovation but may increase compliance efforts, potentially burdening companies.
3. Hybrid models offer balance between user protection and operational feasibility.
4. Enforcement remains challenging, particularly as AI evolves rapidly and generates new ethical and psychological risk.

TABLE II. COMPARATIVE GLOBAL AI CHATBOT REGULATIONS

Country/Region	***Primary Focus***	***Key Features***	***Enforcement/Notes***
EU (GDPR)	Individual rights & data privacy	Consent, data minimization limits, transparency	Strong enforcement, fines for noncompliance
China	State oversight & content control	Real-name verification, age restrictions, content filtering	High operational burden for companies
USA	Consumer protection/risk-based	Age verification, crisis protocols, state-specific oversight	FTC enforcement, state-level variation
South Korea	Hybrid: rights + oversight	Risk assessments, disclosure, user appeal, human review	Pending AI Basic Act, PIPA enforcement
Canada & Brazil	Data protection & AI alignment	Privacy frameworks aligned with GDPR, AI-specific proposals ongoing	Moderate enforcement
Singapore	Innovation + voluntary ethics	Ethical guidelines, soft governance	Reliant on corporate responsibility
India	Minimal regulation	Legacy IT and DPDP Act; non-binding AI guidelines	Limited enforcement

VII. Conclusion

AI chatbots are rapidly transforming communication and knowledge access worldwide, offering substantial benefits while also introducing significant ethical, psychological, and regulatory challenges. This paper examined major risks—including discrimination, hallucination, deception, and psychological manipulation—showing how these emerge from biased data, model limitations, and the persuasive, personalized dynamics of conversational AI.

We also compared global regulatory responses, highlighting how governments differ in their priorities: the EU emphasizes rights and transparency, China centers content control and state security, the United States relies on sector-specific and state-level measures, and countries such as South Korea and Brazil adopt hybrid frameworks. Jurisdictions, like India, with minimal regulation face additional challenges in ensuring user protection and accountability. Together, these differences demonstrate that no single model guarantees safe deployment; effective governance requires balancing innovation, economic utility, and the protection of individual rights.

There remain several important directions for future work. Quantitative risk metrics for chatbot manipulation or emotional dependency would enable more systematic safety evaluation. Automated tools to detect harmful conversational patterns—such as grooming-like behaviors or psychological nudging—could help mitigate manipulation risks. Further study is also needed on how global companies can navigate overlapping regulatory regimes and on the long-term effects of AI companions on minors. Finally, engineering research into benchmarking and reducing hallucination rates remains essential. These avenues together outline a substantial and impactful roadmap for future technical and policy research on AI chatbots.

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